Arbitrarily Long Finite Strings of Growing Returns

Collatz proof project

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Abstract

For every prescribed number of visits, we construct a positive natural Collatz seed whose next consecutive returns to two modulo nine all grow. Each return takes four steps with parity word 1101. This rules out a uniform fixed-return-count descent argument. The seed depends on the requested length; no infinite escaping orbit is constructed.

Use the shortcut map $T(n) = n/2$ for even n and $(3n + 1)/2$ for odd n.

Theorem 1. *For every $R \geq 0$ there is a positive $n \equiv 2 \pmod 9$ such that its first R positive-time returns to that progression occur at times $4, 8, \ldots, 4R$, and every return is larger than the preceding one. There are exactly $3R$ odd steps through time $4R$.*

Proof. If $n = 16q + 11$, exact cylinder transport gives the four iterates

$$24q + 17, \quad 36q + 26, \quad 18q + 13, \quad 27q + 20.$$

The four steps have parity word 1101, and $16T^4(n) = 27n + 23$. If also $n \equiv 2 \pmod 9$, then $q \equiv 0 \pmod 9$; the residues are $8, 8, 4, 2$. Thus this is a first return and strictly increases n.

If 16^{R+1} divides $11n + 23$, reduction modulo sixteen forces $n \equiv 11 \pmod{16}$. The four-step identity gives

$$16\big(11T^4(n) + 23\big) = 27(11n + 23),$$

so 16^R divides $11T^4(n) + 23$. Induction therefore realizes R successive blocks whenever $16^R \mid 11n + 23$ and $n \equiv 2 \pmod 9$.

Define $n_0 = 2$ and $n_{R+1} = 144n_R + 299$. Induction proves

$$11n_R + 23 = 45 \cdot 144^R = 16^R \cdot 45 \cdot 9^R, \qquad n_R \equiv 2 \pmod 9.$$

These positive seeds meet the required conditions. Adding the odd counts of the blocks proves the final count $3R$. ☐

Consequence for the proof program. The coefficient through R returns is $(27/16)^R$, not a contraction. Thus no fixed number of returns guarantees descent uniformly over all positive seeds in this progression. This is compatible with the earlier sixteen-step bound for a *contracting* first return: none of these returns contract. It is also compatible with eventual contraction at a seed-dependent time. The quantifiers are $\forall R\, \exists n_R$, not $\exists n\, \forall R$; this construction does not prove divergence.

Formal verification. `Collatz.GrowingReturns.arbitrarily_many_growing_returns` includes the positive seed, final residue, total odd count, strict growth at every return, and all intermediate residues. Its proof uses exact arithmetic and cylinder transport, with no finite scan or external axiom. The explicit recurrence is `growingReturnSeed`. No mathematical priority claim is made. The remaining program requires a seed-dependent arithmetic argument; fixed powers of this return map cannot supply one.